

Central nervous system The Meninges

Definition: CT membranes that protect brain and spinal cord.

Include:

1. Dura mater (hard mother)

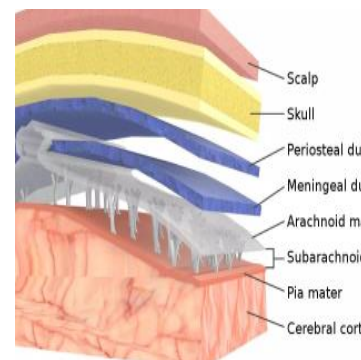
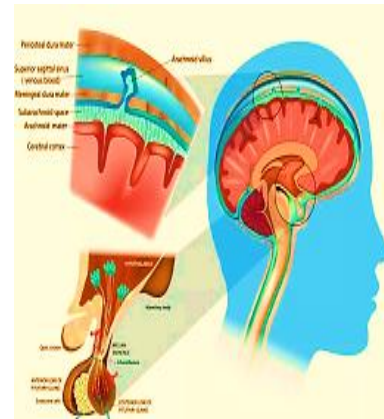
Outer most layer, composed of dense fibro elastic layer strongly adhered to the periosteum of skull.

2. Arachnoid (cobweb like): Two components:

- A. Layer contact with dura (simple squamous epithelium).
- B. Web-like strands of CT like pillars of collagen & elastic fibers holding the roof. The subarachnoid space contains CSF, arteries & veins pass in it. All surfaces of arachnoid are covered by simple squamous epithelium.

3. Pia mater (tender mother)

Innermost vascular loose CT. composed of delicate membrane adheres to the surface of the brain and spinal cord, conveying vessels to them.

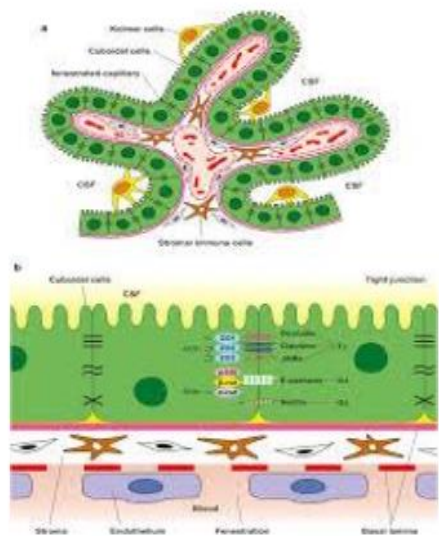


Choroid Plexus

- In the roofs of third and fourth ventricles and in walls of lateral ventricles.
- Consists of invaginated folds of pia mater that penetrates interior of ventricles.
- rich in dilated fenestrated capillaries and covered with ependymal cells.

Ependymal cells: simple cuboidal cells with apical microvilli, basal folds with mitochondria and interdigitating lateral borders (typical ion transporting cells). Basement membrane separates it from the pia mater with wide fenestrated capillaries.

CSF is produced at a constant rate; it completely fills the ventricles, central canal of the spinal cord, subarachnoid space. Excess fluid passes to CSF in the subarachnoid space → superior sagittal venous sinus via finger like projections called arachnoid villi.



NB: No lymphatics in the CNS.

Brain Barriers

A. Blood Brain Barrier:

Barrier between blood and nerve cells.
Prevents the passage of some substances, such as antibiotics, chemicals and bacterial toxic matter, from blood to the nerve tissue.

Structure:

- Tight junctions between non fenestrated endothelium lining capillaries.
- Thick and continuous BM.
- Astrocyte End-Feet: outside BM.

B. Blood CSF Barrier:

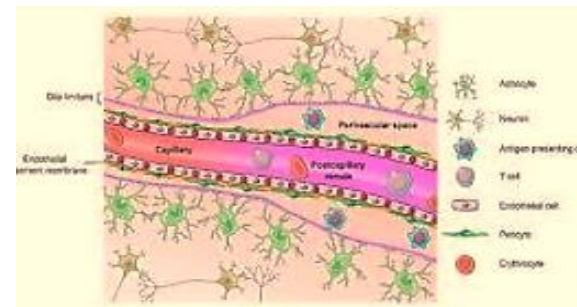
Separates blood from CSF.

Structure:

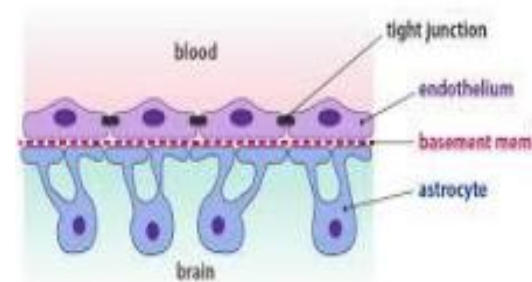
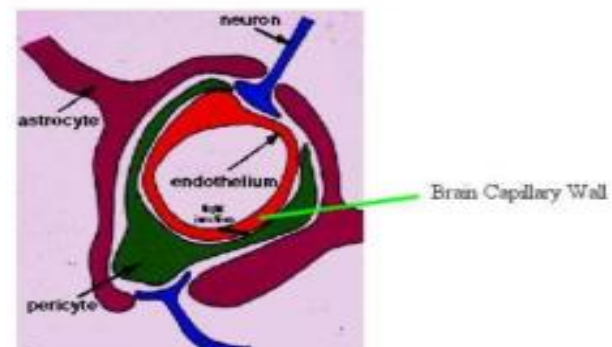
- Tight junctions between fenestrated endothelium lining capillaries.
- BM of fenestrated endothelium.
- BM of ependymal cells
- Ependymal cells.

C. Brain CSF Barrier:

- Ependymal cells.
- Basement membrane.
- Astrocyte End-Feet



Blood brain barrier



Cerebral & Cerebellar cortex

A. Brain:

1. **Cerebrum:** composed of 2 cerebral hemispheres.
2. **Cerebellum:** composed of 2 cerebellar hemispheres.